

Lava REST API Reference

The Lava JSON API is served by the **lava-api-go** service (Go / Gin), the sole API implementation since the legacy Kotlin/Ktor proxy was removed (`build_and_release.sh:16`, `docker-compose.yml:2`). It proxies and structures content from `rutracker.org` (and, on the `/v1/{provider}/...` surface, additional providers) and applies response caching, audit logging, and per-IP rate limiting at the API tier.

This page is derived from:

- `lava-api-go/api/openapi.yaml` — the spec-first wire contract (`info.version: 2.0.0`), source of truth for the **legacy / unprefixd** route group and every DTO schema.
- `lava-api-go/internal/router/router.go` — the single production Gin engine construction (middleware chain + route registration).
- `lava-api-go/internal/handlers/handlers.go` — the unprefixd handler registration.
- `lava-api-go/internal/handlers/v1/handlers.go` — the `/v1/{provider}/...` route group.
- `lava-api-go/internal/auth/middleware.go` — the Lava-Auth HMAC gate.

Scope note. The OpenAPI document at `api/openapi.yaml` describes only the **unprefixd** route group and was authored against the legacy Ktor proxy's wire shape (the two backends were required to be byte-equivalent; see `tests/parity/`). The router additionally mounts a `/v1/{provider}/...` provider-agnostic group (`router.go:102-105`) and the `/health` + `/ready` probes (`router.go:69-70`) that are **not** in `openapi.yaml`. Both surfaces are documented below; where a fact is not backed by a source file it is marked `UNCONFIRMED:`.

1. Servers and transport

From `api/openapi.yaml` `servers:` and `docker-compose.yml`:

URL	Transport	Notes
<code>https://localhost:8443</code>	HTTP/3 (QUIC) +	Default Go-service listener (<code>LAVA_API_LISTEN: ":8443"</code> , <code>docker-compose.yml:66</code>). The

URL	Transport	Notes
	HTTP/2-over-TLS	container exposes 8443/udp (QUIC) and 8443/tcp (Dockerfile:52).

mDNS service type for the Go service is `_lava-api._tcp` (api/openapi.yaml info.description; advertised via `discovery.Announce(...)`, `cmd/lava-api-go/main.go:189`).

A separate metrics listener binds `:9091` (`LAVA_API_METRICS_LISTEN`, `docker-compose.yml:73`) — Prometheus scrape endpoint, not part of this API.

2. Authentication

There are **two distinct authentication concepts** in this API. Do not conflate them.

2.1 Lava-Auth HMAC client-attestation header (gates the whole API)

Source: `lava-api-go/internal/auth/middleware.go`, wired in `router.go:82-85`.

Every request to the **business** routes (everything except `/health` and `/ready`) must carry a client-attestation header. The header **name is not hardcoded** — it is read from `cfg.AuthFieldName` (`LAVA_AUTH_FIELD_NAME`, `docker-compose.yml:77`), per the §6.R no-hardcoding mandate.

The header **value** is:

```
base64( UUID-blob )           # the raw bytes the client sends on
the wire
```

The server decodes the base64 (`middleware.go:61`), then computes `hex( HMAC-SHA256( blob, LAVA_AUTH_HMAC_SECRET ) )` (`hashUUIDBlob`, `middleware.go:118-122`) and matches the result against the configured **active** and **retired** client allowlists (`LAVA_AUTH_ACTIVE_CLIENTS` / `LAVA_AUTH_RETIRED_CLIENTS`, `docker-compose.yml:81-82`). The lookup is constant-time (`constantTimeMapLookup`, `middleware.go:101`).

Outcomes (`middleware.go:55-94`):

Condition	HTTP status	Body
		<code>{"error":"unauthorized"}</code>

Condition	HTTP status	Body
Header absent (hdr == "")	401 Unauthorized	
Header present but not valid base64 / empty blob	401 Unauthorized (backoff counter advances)	{"error": "unauthorized"}
Hash in active allowlist	passes to handler	— (sets client_name, resets backoff)
Hash in retired allowlist	426 Upgrade Required (no backoff advance)	{"error": "client_version_unsupported", "client_name": ...}
Unknown hash	401 Unauthorized (backoff counter advances)	{"error": "unauthorized"}

A backoff/ladder middleware (auth.NewBackoffMiddleware, router.go:83) runs **in front** of the auth middleware; IPs that accumulate failures are short-circuited with 429 + Retry-After before the auth middleware can advance the counter again (router.go:78-84).

The minimum supported client version returned in the 426 body comes from LAVA_AUTH_MIN_SUPPORTED_VERSION_NAME / _CODE (docker-compose.yml:78-79, defaults 1.2.6 / 1026).

2.2 Auth-Token upstream session header (per-request rutracker cookie)

Source: api/openapi.yaml components.parameters.AuthToken.

Auth-Token is an **optional** request header carrying the opaque rutracker session identifier, forwarded as a cookie to the upstream tracker. Absent/empty is permitted; routes that genuinely require an authenticated upstream session (favorites, comment posting, non-public downloads) will return an upstream error or empty result when it is missing.

Auth-Token is the *upstream* credential; Lava-Auth is the *API-tier* client attestation. A request needs Lava-Auth to reach a handler at all, and Auth-Token only to satisfy rutracker for user-scoped routes.

3. Endpoint catalogue — unprefixed routes

Registered in `internal/handlers/handlers.go:105-139`; documented in `api/openapi.yaml` paths:. All routes accept the optional Auth-Token header and require Lava-Auth (§2.1). Status codes below are from the OpenAPI responses: blocks plus the handler error mapper (`writeUpstreamError`, `handlers.go:177-196`).

The upstream-error mapping (`handlers.go:184-195`) is: `ErrNotFound` → 404, `ErrForbidden` → 403, `ErrUnauthorized` → 401, `ErrCircuitOpen` → 503, anything else → 502. Error response bodies are `{}` for parity with the Ktor proxy's `StatusPages` plugin (`handlers.go:178-183`).

Method	Path	operationId	Purpose	Success body
GET	/	getRoot	Health probe — returns <code>checkAuthorized(token)</code>	JSON boolean
GET	/index	getIndex	Alias of /	JSON boolean
POST	/login	postLogin	Log in to rutracker	<code>AuthResponseDto</code>
GET	/forum	getForum	Top-level forum tree	<code>ForumDto</code>
GET	/forum/{id}	getCategoryPage	Paginated category page	<code>CategoryPageDto</code>
GET	/search	getSearch	Search torrents	<code>SearchPageDto</code>
GET	/topic/{id}	getTopic	Topic page (polymorphic)	<code>ForumTopicDto</code>
GET	/topic2/{id}	getTopicPage	Topic page (modern shape)	<code>TopicPageDto</code>
GET	/comments/{id}	getCommentsPage	Paginated comments	<code>CommentsPageDto</code>
POST	/comments/{id}/add	postComment	Submit a comment	JSON boolean
GET	/torrent/{id}	getTorrent	Torrent metadata	<code>TorrentDto</code>
GET	/download/{id}	getDownload	Download <code>.torrent</code> (binary)	<code>application/octet-stream</code>
GET	/captcha/{path}	getCaptcha	Fetch captcha image	image bytes

Method	Path	operationId	Purpose	Success body
GET	/favorites	getFavorites	User's favorited topics	FavoritesDto
POST	/favorites/add/{id}	postFavoritesAdd	Add to favorites	JSON boolean
POST	/favorites/remove/{id}	postFavoritesRemove	Remove from favorites	JSON boolean

Per-endpoint parameters

(From api/openapi.yaml paths:.)

- **GET /** , **GET /index** — header Auth-Token (optional). Returns a JSON boolean from `NetworkApi.checkAuthorized(token)`; absent/empty token typically yields false.
- **POST /login** — application/x-www-form-urlencoded body. Required: username, password. Optional captcha fields: cap_sid, cap_code, cap_val. Never cached. Response is a polymorphic `AuthResponseDto` (discriminator type: Success | WrongCredits | CaptchaRequired | ServiceUnavailable). 400 when required form params are missing.
- **GET /forum** — header Auth-Token (optional, not consumed by the legacy Ktor handler).
- **GET /forum/{id}** — path id (rutracker category id); query page (1-based int32, omitted → first page). 404 for unknown id.
- **GET /search** — query (all optional): query, categories (comma-separated ids), author, authorId, sort (`SearchSortTypeDto`: Date/Title/Downloaded/Seeds/Leeches/Size), order (`SearchSortOrderDto`: Ascending/Descending), period (`SearchPeriodDto`: AllTime/Today/LastThreeDays/LastWeek/LastTwoWeeks/LastMonth), page (1-based int32). 400 on invalid enum constant. Enum values must match the Kotlin enum NAMES exactly.
- **GET /topic/{id}** , **GET /topic2/{id}** , **GET /comments/{id}** — path id; query page (1-based int32). 404 for unknown id. /topic returns the polymorphic `ForumTopicDto`; /topic2 returns the simpler `TopicPageDto`; /comments returns `CommentsPageDto`.
- **POST /comments/{id}/add** — path id; body is raw comment text (text/plain, read via `call.receiveText()`). Auth-Token required (anonymous posting rejected by rutracker). Returns JSON boolean.
- **GET /torrent/{id}** — path id. 404 unknown.
- **GET /download/{id}** — path id; Auth-Token required for non-public torrents. Streams the .torrent binary; sets Content-Disposition (e.g. attachment; filename="rutracker_{id}.torrent") and Content-Type (typically application/x-bittorrent). Never cached. 404 unknown/refused.

- **GET /captcha/{path}** — path path is the URL-safe Base64 encoding of the upstream captcha image URL; decoded server-side and fetched verbatim. 400 if not valid URL-safe Base64.
- **GET /favorites** — Auth-Token required (without it rutracker returns an empty bookmarks page).
- **POST /favorites/add/{id}** , **POST /favorites/remove/{id}** — path id; Auth-Token required. Returns JSON boolean. Idempotent in practice.

4. Endpoint catalogue — /v1/ {provider}/... provider-agnostic routes

Registered in `internal/handlers/v1/handlers.go:44-75` under the Gin group `/v1/:provider` (`router.go:102`). These mirror the unprefix routes but are parameterised by a provider id resolved from the path (provider middleware, `internal/middleware/provider.go` per the package doc at `handlers/v1/handlers.go:5-6`). Registered providers (`cmd/lava-api-go/main.go:132-137`): rutracker, nnmclub, kinozal, archiveorg (Internet Archive), gutenber.

UNCONFIRMED: these `/v1/{provider}/...` routes are not described in `api/openapi.yaml`; the request/response shapes are not re-asserted here beyond the route templates below. Read the per-handler files for the wire shape of each.

Method	Path	Route-template constant	Source
GET	<code>/v1/{provider}/search</code>	—	<code>handlers/v1/search.go</code>
GET	<code>/v1/{provider}/browse/{id}</code>	<code>/v1/{provider}/browse/{id}</code>	<code>handlers/v1/browse.go:17</code>
GET	<code>/v1/{provider}/forum</code>	<code>/v1/{provider}/forum</code>	<code>handlers/v1/forum.go:16</code>
GET	<code>/v1/{provider}/topic/{id}</code>	—	<code>handlers/v1/topic.go</code>
GET	<code>/v1/{provider}/torrent/{id}</code>	—	<code>handlers/v1/torrent.go</code>
GET	<code>/v1/{provider}/download/{id}</code>	—	<code>handlers/v1/torrent.go</code>
GET	<code>/v1/{provider}/comments/{id}</code>	<code>/v1/{provider}/comments/{id}</code>	<code>handlers/v1/comments.go:17</code>
POST	<code>/v1/{provider}/comments/{id}/add</code>	<code>/v1/{provider}/comments/{id}/add</code>	<code>handlers/v1/comments.go:18</code>
GET			

Method	Path	Route-template constant	Source
	/v1/{provider}/favorites	/v1/{provider}/favorites	handlers/v1/favorites.go:16
POST	/v1/{provider}/favorites/add/{id}	—	handlers/v1/favorites.go
POST	/v1/{provider}/favorites/remove/{id}	—	handlers/v1/favorites.go
POST	/v1/{provider}/login	—	handlers/v1/login.go
GET	/v1/{provider}/captcha/{path}	/v1/{provider}/captcha/{path}	handlers/v1/captcha.go:9

Provider error sentinels map to HTTP statuses via `writeProviderError` (handlers/v1/handlers.go:121-128+): `ErrNotFound` → 404, `ErrForbidden` → 403, `ErrUnauthorized` → 401 (read the file for the full set).

5. Liveness / readiness probes (not gated by Lava-Auth)

Registered **before** the auth middleware (router.go:69-70) so orchestrator probes (the podman HEALTHCHECK via the healthprobe binary, and the distribute scripts' health wait) can reach them without a Lava-Auth header.

Method	Path	Handler	Purpose
GET	/health	observability.LivenessHandler()	Liveness — process is up
GET	/ready	observability.ReadinessHandler(...)	Readiness — storage prob (observability.ReadinessPr main.go:170)

6. Schemas (DTOs)

All DTO schemas are defined in `api/openapi.yaml` components.schemas. Sealed-class hierarchies are encoded by `kotlinx.serialization` with the default discriminator field type, whose value is the `@SerializedName` of the concrete subclass; the spec mirrors this with `oneOf + discriminator`. Key polymorphic families:

- **AuthResponseDto** — Success (carries UserDto) | WrongCredits (optional CaptchaDto) | CaptchaRequired (optional CaptchaDto) | ServiceUnavailable (carries reason; see note below).
- **ForumTopicDto** — Topic | Torrent | CommentsPage.
- **PostElementDto** — 19 variants (Text, Box, Align, Size, Color, Bold, Italic, Underscore, Crossed, Quote, Code, Spoiler, Image, ImageAligned, Link, List, Hr, Br, PostBr) — the rich-text post-content tree.
- **ColorValue** — Hex | Name.

AuthResponseDtoServiceUnavailable is an Android-client variant. Per the spec's own note (`openapi.yaml:166-181`), the lava-api-go Go server does **not** currently emit it — its rutracker scraper returns structured Success | WrongCredits | CaptchaRequired only. The variant exists in the spec so the Android Kotlin SDK wire shape stays forward-compatible.

Enumerations: `TorrentStatusDto`, `SearchPeriodDto`, `SearchSortOrderDto`, `SearchSortTypeDto`. Object DTOs: `UserDto`, `CaptchaDto`, `CategoryDto`, `SectionDto`, `ForumDto`, `CategoryPageDto`, `AuthorDto`, `PostDto`, `TorrentDescriptionDto`, `TorrentDataDto`, `TopicPageDto`, `TopicPageCommentsDto`, `SearchPageDto`, `FavoritesDto`, `ProfileDto`, `Error`. See `api/openapi.yaml` for the field-level shape of each.

The canonical, browsable form of these schemas is the OpenAPI document itself; the dev-docs compose profile serves it via Swagger UI at `http://127.0.0.1:8081` (`docker-compose.yml:156-165`).

7. Caching, auditing, rate limiting

- **Response caching** — handlers compute a deterministic cache key against the OpenAPI route-template form (e.g. `/forum/{id}`, not the Gin `:id` form; `handlers.go:30-34`) and read-through / write-through a Postgres-backed cache (`lava_api.response_cache`; see [docs/db/schema.md](#)). `/login`, `/download/{id}`, and `captcha` are never cached (per `openapi.yaml`).
- **Request audit** — `lava_api.request_audit` records method/path/IP/ auth-realm-hash/upstream-status/cache-outcome per request.

- **Rate limiting** — per-IP backoff ladder in front of auth (router.go:83); route-class token buckets persisted in `lava_api.rate_limit_bucket`.
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8. Cross-references

- OpenAPI source of truth: [lava-api-go/api/openapi.yaml](#)
- Router construction: [lava-api-go/internal/router/router.go](#)
- Unprefixed handlers: [lava-api-go/internal/handlers/](#)
- Provider routes: [lava-api-go/internal/handlers/v1/](#)
- Auth middleware: [lava-api-go/internal/auth/middleware.go](#)
- Database schema: [docs/db/schema.md](#)
- Deployment: [docs/deployment/README.md](#)